

standards **Wideband CDMA** and **CDMA2000** are selected as an interfaces multiple access techniques for the European-backed UMTS and IMTS-2000 systems respectively.

2. 5G networks currently offer true data speeds up to 28 Kbps. In comparison, the theoretical speed of 3G can be up to 2 Mbps, that is, approximately 200 times faster than previous 2G networks. This added speed and throughput will make it possible to run applications such as streaming video clips. It is anticipated that 4G speeds could be as high as 100 Mbps. At present, it is important for industries to develop a strong 3G offering that is palatable for the general public. Equally as important, industries must ensure that expectations are realistic and that services meet and exceed those expectations. The cellular evolution path from 1G to 3G and their technologies are discussed in this Chapter.

21.2 First generation (1G)

In mobile phone industry, as a general model of 10 years time is being taken for a generation change from one generation to the next generation. The 1G used analogue transmission techniques for traffic, which was almost entirely voice. It used FDMA technology to achieve radio communications. With the FDMA, the voice channels are carried by different radio frequencies. The technology and features of 1G are shown in Table 21.1.

21.2.1 1G standards

The most successful examples of 1G standards were *Nordic Mobile Telephone (NMT)*, *Total Access Communications System (TACS)*, and *Advanced Mobile Phone Service (AMPS)*. For example, a total of 50 MHz in the band 824–849 MHz and 869–894 MHz is allocated for AMPS. This spectrum is divided into 832 frequency channels (416 downlinks and 416 uplinks).

Drawbacks in 1G: Analogue modulation is sensitive to interference from other users in the system, and the voice quality is quite vulnerable to various kinds of noise. International roaming was not possible in the 1G system due to different nations used different frequency bands and schemes. As a consequence of these, other means of capacity improvement such as efficient modulation schemes were sought for the second generation.

21.3 Second generation (2G)

The cellular industry introduced the second generation (2G) of mobile telephony 10 years later. The 2G services were launched in 1992 and since then they have been expanding and evolving continuously. The technology and features of 2G are shown in Table 21.2.

Table 21.1 Technology and features of 1G

Generation	Technology	Multiple access/ duplex	Bandwidth	Features
1G	AMPS, NMT, TACS	FDMA/FDD	9.6 Kbps	• Analog transmission • Voice service only • No data capability

Table 21.2 Technology and features of 2G

Generation	Technology	Multiple access/ duplex	Modulation	Bandwidth	Features
2G	GSM	TDMA/FDD	GMSK	9.6–14.4 Kbps	Digital voice role transmission
	IS-95 based on CDMA	CDMA/FDD	QPSK		Global roaming Circuit switched data capability

The 2G mobile systems use digital radio transmission for traffic. The difference between 1G and 2G is 1G used analogue signalling while the latter used digital signalling. Thus, the boundary line between 1G and 2G systems is obviously the analogue/digital split.

21.3.1 Advantages of digital system use in 2G

- Digital data can be compressed and multiplexed much more effectively than analogue voice encodings.
- Digital cellular systems have many features, such as improved communication quality due to various digital signal processing technologies.
- Digital technology enables the use of signal processing techniques to increase robustness against interference.
- It also reduces the spectral bandwidth required for each user and hence provides higher capacity.
- 2G provides about 3–4 times the capacity of the 1G without adding new base stations. One frequency channel is simultaneously divided among several users (either by code or time division).
- Since digital systems are more immune to noise, a Signal to Interference Ratio (SIR) of 7 dB could be tolerated for a digital system, whereas 15 dB is required for the analogue systems under the same circumstances.
- Digital voice encoding allows for lower powered radio signals that require less battery power. This is possible because of CODEC introduction which encodes and decodes digital data stream or signal (Figure 21.1).
 - The digital voice encoding also allows digital error checking (increases sound quality and lowers the noise level).
 - Going to "all-digital" allowed for the introduction of digital data transfer (SMS—"short message service" and e-mail).

21.3.2 Disadvantages of digital system use in 2G

- Cellular towers had a limited coverage area.
- Abrupt dropped calls due to uneven decay curve.
- Spotty coverage, which is shown in Figure 21.2.

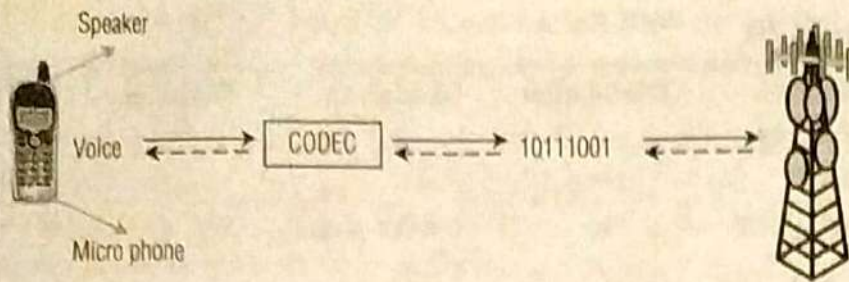


Figure 21.1 Digital voice encoding

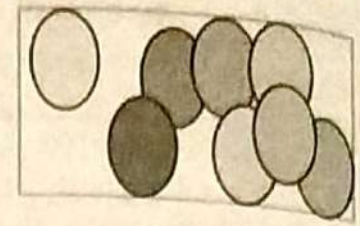


Figure 21.2 Spotty coverage

21.3.3 2G standards

Various 2G cellular systems were developed in the 1990s. There are four main standards for 2G systems:

- Global System for Mobile (GSM) communications and its derivatives.
- Digital AMPS (D-AMPS).
- Code Division Multiple Access (CDMA) IS-95 and
- Personal Digital Cellular (PDC).

The analogue cellular systems (1G) in each country used different frequency bands and schemes, which made interconnection impossible across national borders.

In 1982, the European Conference of Postal and Telecommunications Administrations (CEPT) established the Group Special Mobile, and development efforts were carried out under the leadership of the European Telecommunications Standards Institute (ETSI). GSM-based services were launched in 1992. The 2G standards discussed earlier can be divided into two groups:

- TDMA-based 2G standards.
- IS-95 standard-based CDMA technology.

21.4 TDMA-based 2G standards

Most of the 2G systems employ Time Division Multiple Access (TDMA), such as the Global System for Mobile (GSM) Communications, Digital AMPS (D-AMPS), the Digital Cellular System 1800 (DCS1800) in Europe, the Interim Standard (IS) 54 in the USA, and the Personal Digital Cellular (PDC) system in Japan.

With TDMA, the time axis is subdivided into different non-overlapping time slots where each user is exclusively assigned each time slots in which this user employs the total available bandwidth.

Thus, TDMA separates the users in the time domain. In practice, TDMA is combined with FDMA to reduce the hardware complexity of an otherwise extremely broadband system and to increase the flexibility of the system.

21.4.1 Global system for mobile (GSM) communications

GSM is the most successful and widely used 2G system. Originally designed as a pan-European standard, it was quickly adopted all over the world. GSM is widely used in Europe, Australia, and

Asia. In GSM, every frequency carrier is divided into fixed time slots that support up to eight voice channels. The speech coding rate is 13 Kbps in GSM. With TDMA, the radio hardware in the base station can be shared among multiple users. Only in America, the GSM has not reached a dominant position yet. In North America, Personal Communication System-1900 (PCS-1900; a GSM derivative, also called GSM-1900) has gained some ground, and in South America, Chile has a wide-coverage GSM system. The GSM standard operates at a different set of frequencies (uplink and downlink channels) worldwide and mainly at 850, 900, 1,800, and 1,900 MHz. The forward channel (downlink) and reverse channel (forward link) ranges of GSM 00 are 935–960 MHz and 890–915 MHz, respectively.

21.4.2 Digital AMPS (D-AMPS)

With TDMA, the radio hardware in the base station can be shared among multiple users. In North America, however, the main design objective has been to make a smooth transition from the low-capacity analogue systems to high-capacity digital systems. This is possible since digital technology enables allocation of three TDMA channels on the same radio frequency as one FDMA channel in the AMPS system. Such a mixed system, known as ADC (D-AMPS or IS-54 standard), enhances the capacity of the system three times just by exchanging the analogue FDMA transceivers to digital TDMA transceivers. The speech coding rate is 7.95 Kbps in ADC. PDC and ADC systems have high modulation efficiency due to the use of QPSK modulation and low bit rate codec. Therefore, both systems have more system capacity than GSM.

21.4.3 Personal digital cellular (PDC)

PDC is the Japanese 2G standard. Originally it was known as *Japanese Digital Cellular* (JDC), but the name was changed to *Personal Digital Cellular* (PDC) to make the system more attractive outside Japan. However, this renaming did not bring about the desired result, and this standard is commercially used only in Japan. The specification is known as RCR STD-27, and the system operates in two frequency bands: 800 MHz and 1,500 MHz. It has both analogue and digital modes. Its physical layer parameters are quite similar to D-AMPS, but its protocol stack resembles GSM. PDC has been a very popular system in Japan. This success has also been one of the reasons that the Japanese have been so eager to develop 3G systems as soon as possible, as the PDC system capacity is quickly running out.

21.5 IS-95 or IS-136 (CDMA) or cdmaOne standard)

21.6 Two point five generation (2.5G)

"Generation 2.5" is a designation that includes all advanced upgrades for the 2G networks. The need for increased throughput data rates in data transfer (such as web browsing and e-mail) led to the evolution of 2.5G. These upgrades may in fact sometimes provide almost the same capabilities as the planned 3G systems. The boundary line between 2G and 2.5G is an unclear one. It is difficult to say when a 2G becomes a 2.5G system in a technical sense. In order for the evolution from 2G to 2.5G to be backwards compatible and cost bearable, 2.5G standard includes the following technologies:

- HSCSD
- GPRS
- EDGE
- *Interim Standard 95B (IS-95B)*

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Table 21.3 Technology and features of 2.5G

2.5G		Technology	Modulation	Multiple access	Bandwidth	Features
GSM	HSCSD	High-Speed Circuit-Switched Data	GMSK	TDMA	9.6–57.6 Kbps	Extension of GSM Higher data speeds
	GPRS	General Packet Radio Services	GMSK	TDMA	9.6–115 Kbps	Extension of GSM Always on connectivity Packet-switched data
	EDGE	Enhanced Data Rates for Global Evolution	8-PSK	TDMA	64–384 Kbps	Extension of GSM Always on connectivity Faster than GPRS
	IS-95B or CDMA 2000	High Data Rate (HDR), Code division multiple access	QPSK	CDMA	64 Kbps–2.4 Mbps	Extension of cdma one or IS-95 Faster than 2.5 GSM standard

Generally, a 2.5G GSM system includes at least one of the HSCSD, GPRS, and EDGE technologies. An IS-95 system is called 2.5G when it implements IS-95B, or CDMA2000 1xRTT upgrades. The technology and features associated with 2.5G are illustrated in Table 21.3.

Upgrade of 2G GSM to 2.5 GSM

The biggest problem with the 2G standard GSM is its low air interface data rates. The basic GSM could provide a maximum of 9.6 Kbps to 14.4 Kbps user data rate.

Anyone who has tried to web surf with these rates knows that it can be a rather desperate task. Various methods that are adopted for the upgradation of 2G GSM to 2.5 GSM are given in the following (Figure 21.7).

21.6.1 High-speed circuit-switched data (HSCSD)

HSCSD is the easiest way to speed things up. This means that instead of one time slot, a mobile station can use several time slots for a data connection. In current commercial implementations, the maximum is usually four time slots. One time slot can use either 9.6 Kbps or 14.4 Kbps speeds. The total rate is simply the number of time slots times the data rate of one slot. This is a relatively inexpensive way to upgrade the data capabilities, as it requires only software upgrades to the network (plus, of course, new HSCSD-capable phones), but it has drawbacks. The biggest

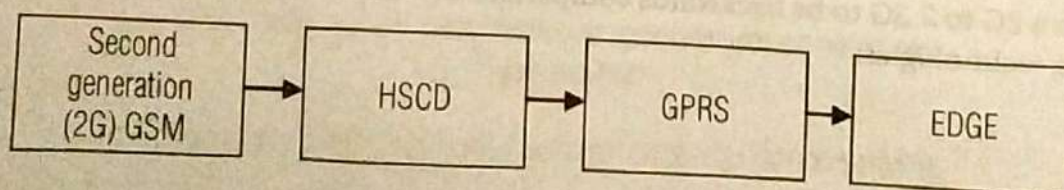


Figure 21.7 Upgrade of 2G GSM to 2.5 GSM

problem is the usage of scarce radio resources, because it is *circuit switched*, HSCSD allocates the used time slots constantly, even when nothing is being transmitted. In contrast, this same feature makes HSCSD a good choice for real-time applications, which allow for only short delays. The high-end users, which would be the most probable HSCSD users, typically employ these services in areas where mobile networks are already congested. Adding HSCSD capability to these networks certainly will not make the situation any better. An additional problem with HSCSD is that handset manufacturers do not seem very interested in implementing HSCSD. It can be seen that HSCSD will be only a temporary solution for mobile data transmission needs.

21.6.2 General packet radio services (GPRS)

The next solution is GPRS. With this technology, the data rates can be pushed up to 115 Kbps, or even higher if one can forget error correction. However, with adequate data protection, the widely quoted 115 Kbps is the theoretical maximum in optimal radio conditions with eight downlink time slots. A good approximation for throughput in "average" conditions is 10 Kbps per time slot. What is even more important than the increased throughput is that GPRS is packet switched, and thus it does not allocate the radio resources continuously but only when there is something to be sent.

The maximum theoretical data rate is achieved when eight time slots are used continuously. The first commercial launches for GPRS took place in 2001. GPRS is especially suitable for non-real-time applications, such as e-mail and web surfing. Also, bursty data is well handled with GPRS, as it can adjust the assigned resources according to current needs. It is not well suited for real-time applications, as the resource allocation in GPRS is contention based; thus, it cannot guarantee an absolute maximum delay.

21.6.3 Enhanced data rates for global evolution (EDGE)

EDGE was designed specifically as an upgrade to GPRS for integration into GSM network starting from the GSM community as a path to 3G. It uses the same basic GSM infrastructure with the difference being that it can use 8-PSK modulation in addition to the GMSK. It has nine different air interfaces called *Multiple Modulation Coding Schemes* (MCS), which is named from 1–9. Every MCS has a varying control protection and can use either GMSK for a low data rate (8.8–17.6 Kbps) or 8-PSK for a high data rate (22.4–59.2 Kbps) for each time slot. According to the level of error correction needed for the application every mobile user can adopt whatever MSC is suitable without the error protection and eight time slots taken when it theoretically connects with $8 \times 59.2 = 547.2$ Kbps. A minimum error control and network considerations limit the throughput at 384 Kbps. It requires new hardware (routers, gateways) and software updates at the base stations. Figure 21.8 presents the worldwide EDGE coverage.

The third 2.5G improvement to GSM is EDGE. Originally this acronym stood for Enhanced Data rates for GSM Evolution, but now it translates into Enhanced Data rates for Global Evolution as the EDGE idea can also be used in systems other than GSM. The idea behind EDGE is a new modulation scheme called *eight-Phase Shift Keying* (8PSK). It increases the data rates of standard GSM up to threefold. EDGE is an attractive upgrade for GSM networks, as it only requires a software upgrade to base stations if the RF amplifiers can handle the non-constant envelope modulation with EDGE's relatively high peak-to-average power ratio. It does not replace but rather coexists with the old *Gaussian Minimum Shift Keying* (GMSK) modulation, so mobile users can continue using their old phones if they do not immediately need the better service quality provided by the higher data rates of EDGE. It is also necessary to keep the old GMSK because 8PSK

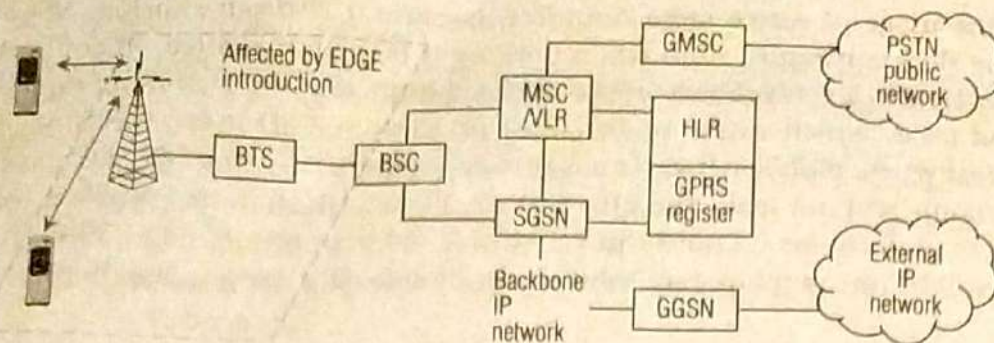


Figure 21.8 GSM/EDGE network structure

can only be used effectively over a short distance. For wide area coverage, GMSK is still needed. If EDGE is used with GPRS, then the combination is known as *enhanced GPRS* (EGPRS). The maximum data rate of EGPRS using eight time slots (and adequate error protection) is 384 Kbps. Note that the much-advertised 384 Kbps is thus only achieved by using all radio resources of a frequency carrier, and even then only when the mobile station is close to the base station. ECSD is the combination of EDGE and HSCSD, and it also provides data rates three times the standard HSCSD. A combination of these three methods provides a powerful system, and it can well match the competition by early 3G networks.

21.6.4 Interim standard 95B (IS-95B)

The IS-95 (CDMA) standard currently provides 14.4 Kbps data rates. It can be upgraded to IS-95B, which is able to transfer 64 Kbps with the use of multiple code channels. Interim Standard 95B (IS-95B) is the revision of IS-95 and IS-95A networks proposed in 1995.

The term **cdmaOne** is used to describe cellular networks based on IS-95A and IS-95B technology. *cdmaOne* provides packet and circuit switched data access through CDMA radio channels.

The first IS-95B networks were originally deployed in September 1999 in Korea by dedicating eight different orthogonal user channels simultaneously, and theoretically it can achieve $8 \times 14.4 = 115.2$ Kbps. Due to this data speed the IS-95B is capable of reaching packet data service, therefore it is categorized as a 2.5G technology.

2G cdmaOne advantages

cdmaOne technology offers numerous benefits to the 2G cellular operators and their subscribers:

- Capacity increases from 8 to 10 times that of an AMPS analogue system and 4 to 5 times that of a GSM system.
- Improved call quality, with better and more consistent sound as compared to AMPS systems.
- Simplified system planning through the use of the same frequency in every sector of every cell.
- Enhanced privacy.
- Improved coverage characteristics, allowing for the possibility of fewer cell sites.
- Increased talk time for portables.
- Bandwidth on demand.

21.7 Third generation (3G) development

Need for 3G cellular systems: The need for high speed Internet access up to 2 Mbps, live video communications, fast web access, and the simultaneous data and voice transmission led to the development of 3G cellular networks. The new services have to be available in indoor and outdoor environments, where the integration of satellite links shall enable world-wide coverage. These requirements cannot be completely covered by the 2G systems that have a relatively low available bit rate in the range of 14.4 Kbps per user, primarily designed for speech transmission applications. Research activities concerning the standardization of the 3G mobile radio systems are in progress world-wide.

IMT-2000/UMTS

The International Telecommunications Union (ITU) created a project in 1998 for a common worldwide cellular standard under the name *International Mobile Telephone 2000 (IMT-2000)*. IMT-2000 is also referred to as *Universal Mobile Telecommunications Systems (UMTS)* in Europe.

The technology and features associated with the 3G are illustrated in Table 21.4.

21.7.1 Features of 3G mobile communications

1. With 3G, the information is split into separate but related packets before being transmitted and reassembled at the receiving end. Packet switched data formats are much more common than their circuit switched counterparts.
2. The World Wide Web (WWW) is becoming the primary communication interface. People access the Internet for entertainment, services, and information collection, the intranet for accessing enterprise information and connecting with colleagues, and the extranet for accessing customers and suppliers. These are all the derivatives of the WWW aimed at connecting different communities of interest. Information and other resources are being stored in remote Web servers, which serve the various needs of human beings through Web browsers at their ends.

Table 21.4 Technology and features of 3G

Generation	Technology	Multiple access/ duplex	Modulation	Bandwidth	Features
3G	IMT-2000 International mobile telecommunication 2000/-	TDMA/FDD	GMSK	64-2,048	Always on Global roaming IP enabled
	UMTS Universal mobile telecommunication	CDMA/FDD	QPSK		

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3. Speeds of up to 2 Mbps are achievable with 3G. The data transmission rates will depend upon the environment; the call is being made in, however, only indoors and in stationary environments that these types of data rates will be available. For high mobility, the data rates of 144 Kbps are expected to be available.

21.8 3G Air interface technologies

Since the 2G cellular infrastructures already had different implementation variations, an important challenge for UMTS and IMT-2000 is the selection of an appropriate *multiple access scheme* to meet the demands of 3G mobile radio systems. The IMT-2000 standard accepts 5 possible radio interfaces based on 3 multiple access technologies (FDMA, TDMA, and CDMA) as illustrated in Figure 21.9. The five 3G air interface technologies are as follows:

- Wideband CDMA (WCDMA)
- Code Division Multiple Access 2000 (cdma2000)
- Time Division Synchronous Code Division Multiple Access (TD-SCDMA)
- Universal Wireless Communications-136 (UWC-136)
- Digital Enhanced Cordless Telephone (DECT)

Among the five air interfaces as shown in Figure 21.9, three air interfaces **WCDMA**, **CDMA2000**, and **TD-SCDMA** are the main interfaces and became popular.

- The first two interfaces **WCDMA** and **CDMA2000** are the variations of CDMA technology and are based on FDD.
- The third interface **TD-SCDMA** was proposed by the China and is based on TDD. It uses TDMA/TDD and CDMA techniques for the high-speed data transfer over GSM networks up to 384 Kbps. It has a 1.6 MHz radio channel and a 5 ms frame subdivided into seven time slots.
- The fourth interface **UWC-136** falls under the TDMA category, which is also known as EDGE.
- The fifth interface falls under the **FD-TDMA** category and is known as **DECT**. The DECT is mainly used in the indoor environments.

From the above explanation, it is clear that the overall infrastructure of 3G mostly includes WCDMA and CDMA techniques in terms of applicability and future potential.

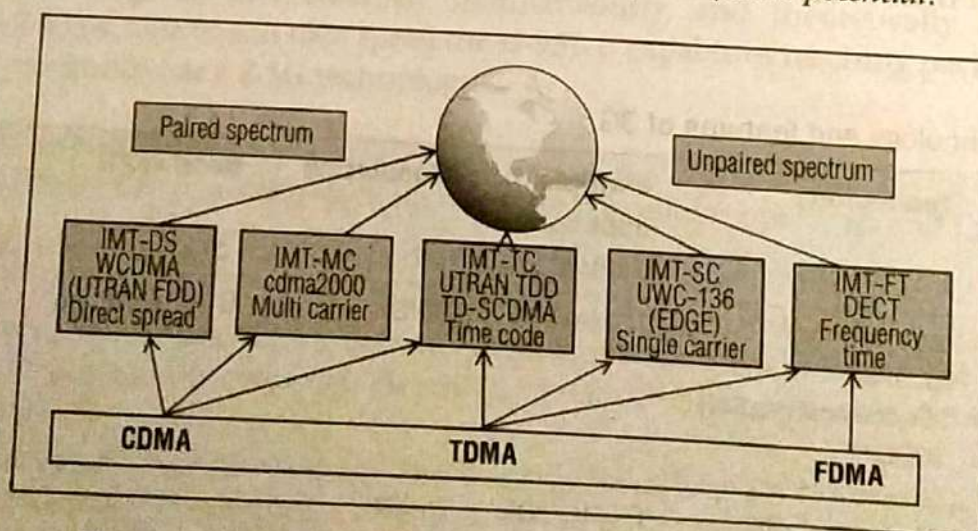


Figure 21.9 Five radio interfaces based on 3 technologies

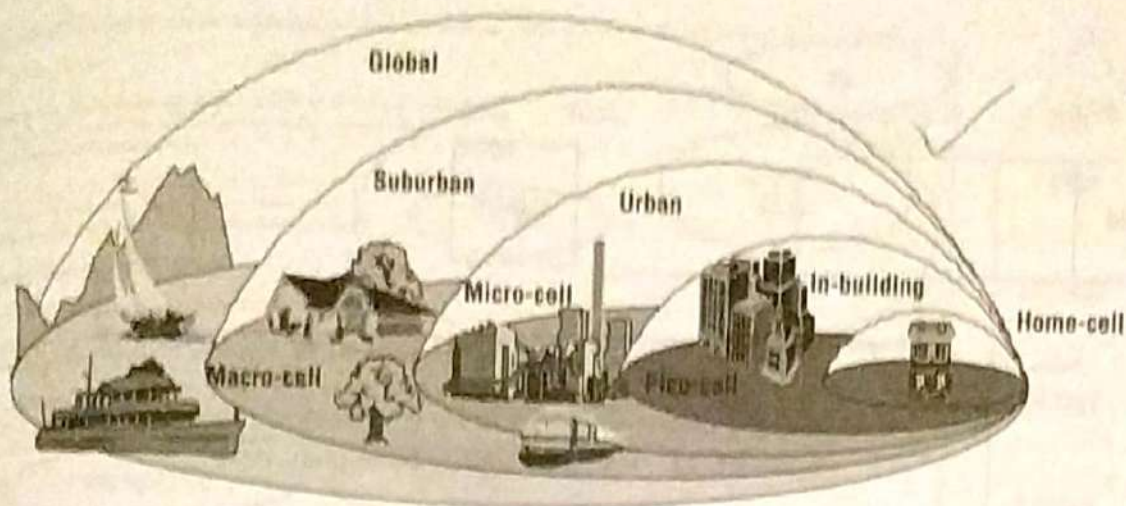


Figure 21.10 Different environments for UMTS

WCDMA-based UMTS will offer a common air interface covering home, office, car, train, aeroplane, or a pedestrian as shown in Figure 21.10. UMTS will integrate all the services offered by different mobile communication systems such as mobile telephone, cordless telephone, public air radio, satellite radio, etc in one service. It will allow users to roam during an existing connection between different types of communication networks. UMTS will offer broadband services, that is, it will be possible to transmit voice, text, data, and images over one connection.

21.9 3G spectrum

The channel spacings (bandwidths) used for making a call in 1G and 2G communication services are 30 kHz and 30–200 kHz, respectively. The amount of bandwidth needed for 3G services is 15–20 MHz, and you can see that there is as much as a 500 times increase in the amount of bandwidth required.

The 3G spectrum was first identified at World Administrative Radio Congress (WARC-92). Worldwide frequency spectrum identified for the 3G transmission is in the 1,885–2,025 MHz and 2,110–2,200 MHz bands. Of these 230 MHz of 3G spectrum, 2×30 MHz were intended for the satellite component of IMT-2000 and the rest for the terrestrial component. The main IMT-2000 standardization effort was to create a new air interface that would increase frequency usage efficiently. The 3G also identifies the paired and unpaired parts of the spectrum.

For example, in the IMT-2000 3G spectrum, the satellite service uses the bands 1,980–2,010 MHz for uplink and 2,170–2,200 MHz for downlink. This leaves the three bands 1,900–1,980 MHz, 2,010–2,025 MHz, and 2,110–2,170 MHz for terrestrial component.

Paired spectrum: Radio spectrum is often organized (and sold) as *paired spectrum* with a bit of spectrum in a lower frequency band and a bit of spectrum in an upper frequency band. Paired spectrum is often specified in a form like " 2×15 MHz" meaning 15 MHz in a lower band and 15 MHz in an upper band. For the paired spectrum, the bands for uplink (mobile transmit) and downlink (base-station transmit) are identified for *Frequency Division Duplex* (FDD) operation.

The technique of two users talking to each other on two separate frequencies is called frequency Division Duplex (FDD). WCDMA is an FDD technique.

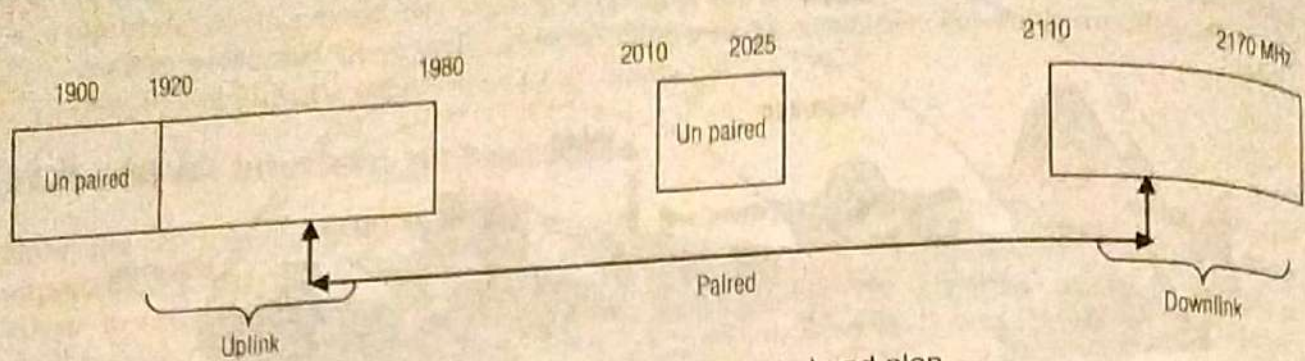


Figure 21.11 UMTS frequency band plan

Unpaired spectrum: The unpaired bands, for example, used for *Time Division Duplex* (TDD) operation. TD-CDMA is a TDD technique. In this duplex method, uplink and downlink transmissions are carried over by the same frequency band by using synchronized time intervals. Thus, time slots in a physical channel are divided into transmission and reception part. Note: The band that is most globally deployed for 3G is still 2 GHz.

The WCDMA air interface was selected for paired frequency bands (FDD operation) and TDCDMA (TDD operation) for unpaired spectrum.

Then, somewhat diverging arrangements are made between regions of the frequency bands assigned to 3G means that there is not even a single band that can be used for 3G roaming worldwide. The choice of frequency band for implementing in Europe and Asia is clear; for example, the UMTS. The 3G spectrum plan in Europe is shown in Figure 21.11.

We can observe from the earlier discussion that the UMTS FDD is designed to operate in paired bands, with uplink in the 1,920–1,980 MHz and downlink in the 2,110–2,170 MHz band. UMTS TDD was operated with the unpaired frequency bands 1,900–1,920 MHz and 2,010–2,025 MHz.

In case of the United States, these frequency bands are not available; they have already using these frequencies. Therefore, the three frequency bands suggested for implementation of UMTS in the United States are

- 806–890 MHz band
- 1,710–1,885 MHz band
- 2,500–2,690 MHz

21.10 Internet speeds of 2G, 2.5G, and 3G technologies

2.5G is the interim solution for the current 2G networks to have 3G functionality. 2.5G networks are being designed such that a smooth transition (software upgrade) to 3G can be realized. 2.5G networks currently offer true data speeds up to 28 Kbps. In comparison, the theoretical speed of 3G can be up to 2 Mbps, that is, approximately 200 times faster than previous 2G networks. This added speed and throughput will make it possible to run applications such as streaming video clips. It is anticipated that 4G speeds could be as high as 100 Mbps. Thus, 4G will represent another quantum leap in mobile Internet speeds and picture quality. Ericsson confirms that 4G

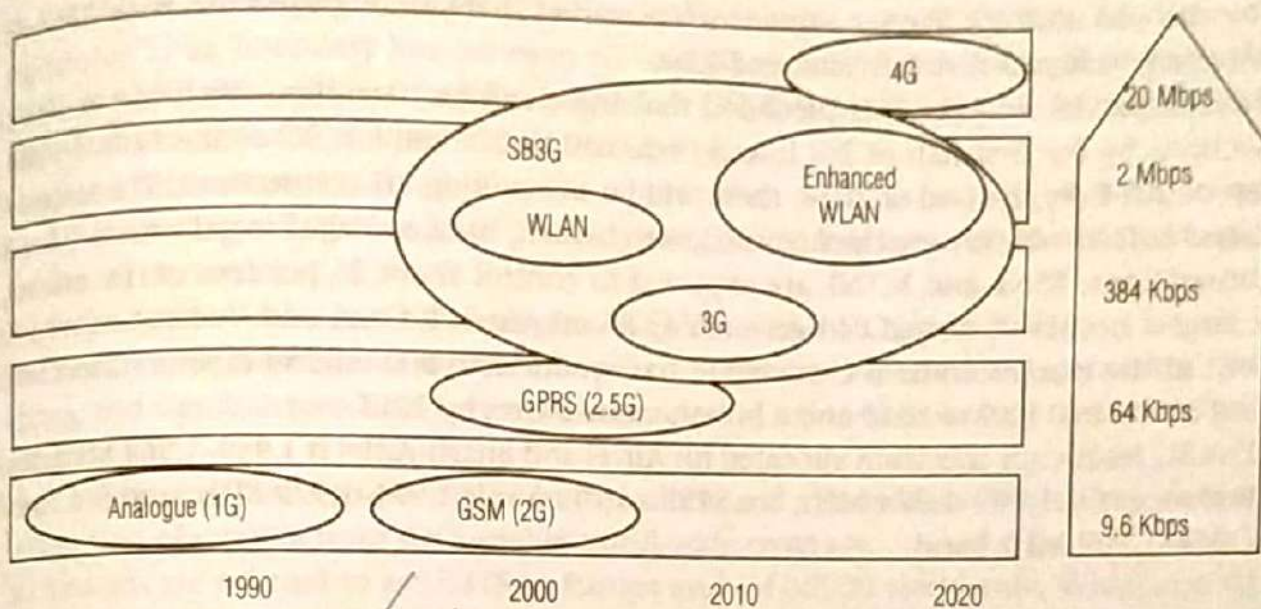


Figure 21.12 Mobile generation

could bring connection speeds of up to 50 times faster than 3G networks and could offer three-dimensional visual experiences for the first time. Figure 21.12 represents the typical progression of wireless communications.

21.11 Limitations of 3G

3G performances may not be sufficient to meet the needs of future high-performance applications like multimedia, full-motion video, and wireless teleconferencing. We need a network technology that extends 3G capacities by an order of magnitude.

1. There are multiple standards for 3G, making it difficult to roam and interoperate across networks. We need global mobility and service portability.
2. 3G systems are based on primarily a wide-area concept. We need hybrid networks that utilize both wireless LAN (hot spot) concept and cell or base-station wide area network design.
3. We need wider bandwidth.
4. Researchers have come up with spectrally more efficient modulation schemes that cannot be retrofitted into 3G infrastructures.
5. We need all digital packet networks that utilizes IP in its fullest form with converged voice and data capability.

23.2 4G evolution

The existence of several diverse 3G standards (e.g., WCDMA and CDMA2000) limits seamless *global roaming* between different cellular networks for a mobile user with a single handset. In addition, there is a fundamental difference between wireless cellular networks (1G, 2G, or 3G) and wireless data networks such as WLANs and PANs.

The difference between wireless cellular networks (1G, 2G, or 3G) and **wireless data networks** (WLANS, PANS) is that wireless cellular systems are circuit-switched while wireless data networks are packet-switched.

As mentioned in the previous chapter, all over the world, the majority of mobile communication service providers are operating the networks using two different families of standards:

- **3GPP** based (i.e., GSM, EDGE, UMTS, HSDPA, HSPA+)
- **3GPP2** based (i.e., CDMA IS-9, CDMA IS-2000, 1x Data Only Revision 0 (1xDO Rev0), 1x Data Only Revision A (1xDO RevA))

Both families of standards were set to evolve to separate 4G technologies:

- 3GPP family to LTE
- 3GPP-2 family to Ultra Mobile Broadband (UMB)

In addition, a brand-new wireless communication standard, IEEE802-16e1, was proposed by IEEE in 2005. Service providers have been participating in the development of LTE and have committed to adopting the LTE technology. With no adoption, work on UMB has been halted and mobile WiMAX has seen very little adoption by mobile communication service providers. Therefore, there is an ongoing convergence of mobile communication standards after many years of diverse standards deployed in various geographical locations (most of the North American networks are based on the 3GPP2 standards and Europe has deployment of networks that are based on the 3GPP standards). Both Americans and Europeans have continuous commitment to the adoption of a single standard (i.e., LTE) by both 3GPP-based and non-3GPP-based service providers.

Convergence issues for these differences between the wireless cellular systems and the wireless data networks will be addressed in the design of 4G cellular networks. It is envisioned that 4G networks, possibly running on a common IP-based backbone, will provide users with seamless wireless access to voice, data, and video services, irrespective of which wireless network they belong to. The fundamental reason for the transition to the all-IP is to have a common platform for all the technologies that have been developed so far, and to harmonize with user expectations of the many services to be provided.

The data rates of 100–155 Mbps (or speeds) are expected in the 4G cellular and to handle these data rate, OFDM, and among the other proposed technologies, UWB and MIMO antenna systems are emerging as the access method of choice. OFDM is spectrally more efficient than existing access methods such as TDMA, FDMA, CDMA, WCDMA, or CDMA2000. We focus on UWB radio due to its unique advantages: high data rate, low power, and resilience to multipath fading effects. Table 23.1 illustrates a timeline for 1G to 4G cellular systems.

OFDM is a successor to frequency hopping and direct sequence CDMA. It has the capability to cancel multipath distortion in a spectrally efficient manner without requiring multiple local oscillators. It uses the principles of IFFT and FFT. Also OFDM uses frequency orthogonality as compared to code orthogonality used in CDMA.

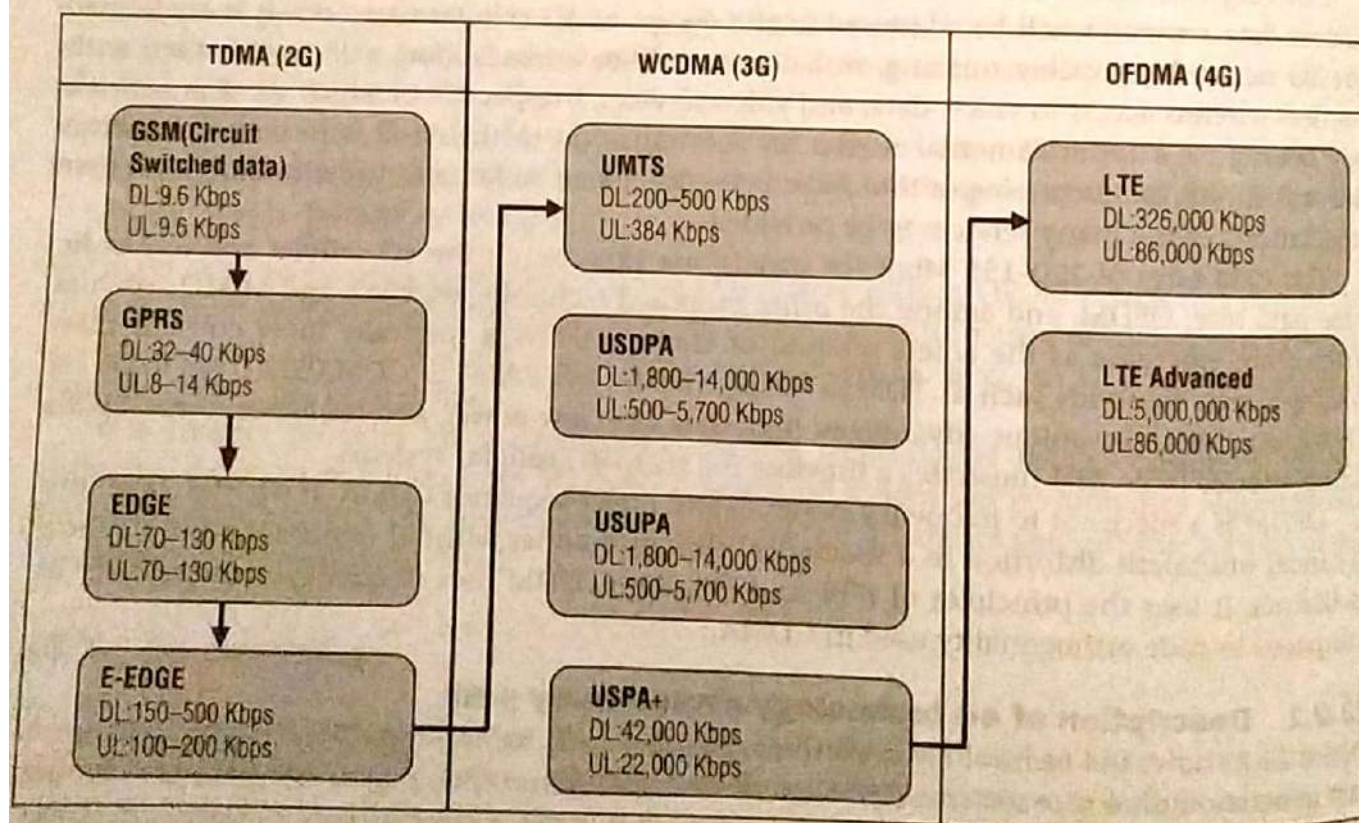
23.2.1 Description of 4G technology evolutionary path

Figure 23.2 shows the technology evolutionary path for 4G technology. The development from first generation analogue systems (1985) to second generation (2G) digital GSM (1992) was the heart of the digital revolution. But much more than this, it was a huge success for standardization emanating from Europe and gradually spreading globally.

Table 23.1 A timeline for 1G to 4G cellular systems

Technology	1G	2G	3G	4G
Design began	1970	1980	1990	2000
Implemented	1983	1991	2003	?
Standards	AMPS, TACS NMT	GSM,CDMA,PDC 2.5G standards are: GPRS, EDGE, 1xRTT, IS95B	WCDMA, CDMA2000	One standard
Service	Analog voice	Digital voice Short messages	Higher capacity Broadband	Higher capacity, Complete IP Oriented, multimedia
Data bandwidth	1 Kbps	14.4 Kbps	2 Mbps	1,000 Mbps
Multiplexing	FDMA	TDMA,CDMA,	CDMA	OFDM

However, world-wide roaming still presents some problems with standards IS-95 (a CDMA rather than a TDMA digital system) and IS-136 (a TDMA variant) still entrenched in some countries. Extensions to GSM (2G) via General Packet Radio Service (GPRS) and EDGE (E-GPRS), as well as WAP and i-mode (so called 2.5G), will allow the transmission of higher data rates as well as speech prior to the introduction of 3G.


Figure 23.2 Evolutionary path for 4G technology

GSM was designed for digital speech services or for low bit rate data that could fit into a speech channel (e.g., 9.6 Kbps). It is a *circuit* rather than a packet-oriented network and hence is inefficient for data communications. To address the rapid popularity increase of Internet services, GPRS is being added to GSM to allow packet IP communications at up to about 100 Kbps.

3G systems were standardized in 1999. These include International Mobile Telecommunications 2000 (IMT-2000), which was standardized within ITU-R and includes the Universal Mobile Telecommunications System (UMTS) European standard from European Telecommunications Standards Institute (ETSI), the US-derived CDMA2000 and the Japanese NTT DoCoMo Wideband Code Division Multiple Access (W-CDMA) system. Such systems extend services to (multi-rate) high-quality multimedia and to convergent networks of fixed, cellular, and satellite components.

The radio air interface standards are based upon W-CDMA (UTRA FDD and UTRA TDD in UMTS, multicarrier CDMA2000, and single carrier UWC-136 on derived US standards). The core network has not been standardized, but a group of three evolved GSM (MAP), evolved ANSI-41 (from the American National Standards Institute), and IP-based are all candidates. 3G is also about a diversity of terminal types, including many non-voice terminals such as those embedded in all sorts of consumer products. Bluetooth (another standard not within the 3G orbit, but likely to be associated with it) is a short-range system that addresses such applications. Thus, services from a few bits per second up to 2 Mbps can be envisioned.

Long-term Evolution (LTE): LTE is an emerging technology for higher data rates. It is also referred to as 3.9G or super 3G technology. LTE is developed as an improvement to Universal Mobile Telecommunication System by 3rd Generation Partnership Project (3GPP). LTE uses Orthogonal Frequency Division Multiple Access (OFDMA). The download rate in LTE is 150 Mbps and it utilizes the available spectrum in a very sophisticated way. In LTE, the IP packet delay is less than 5 ms which provides the experience of wired broadband Internet access in wireless environment. The mobile TV broadcast is facilitated by LTE over LTE network.

23.3 Objectives of the projected 4G

4G will be a fully IP-based integrated system. This will be achieved after wired and wireless technologies converge and will be capable of providing between 100 Mbps and 1 Gbps speeds, both indoors and outdoors, with quality and high security. 4G will offer all types of services at an affordable cost. The following are the objectives of the 4G wireless communication standard:

- A spectrally efficient system (in bps/Hz and bps/Hz/site).
- High network capacity: more simultaneous users per cell.
- 4G systems support streaming video, voice calls, Internet, and many more broadband services.
- A nominal data rate of 100 Mbps while the client physically moves at high speeds relative to the station, and 1 Gbps while client and station are in relatively fixed positions.
- Smooth handoff across heterogeneous networks.
- Seamless connectivity and global roaming across multiple networks.
- High QoS for next generation multimedia support (real-time audio, high-speed data, HDTV video content, mobile TV, etc).
- Interoperability with existing wireless standards.
- IP-based mobile technology.

23.4 Advantages of 4G network technology over 3G

With the 3G network deployment yet to pick up speed, 4G technology is already in view. If the predictions of the mobile industry experts prove to be true, 4G network deployment may start anytime in this coming decade. Trials are already being conducted by some mobile operators and vendors. Table 23.2 gives a comparison of few key features of 3G and 4G technologies.

The need for the 4G technology is that the present 3G capability is considered to be substantially less than predicted future requirements and applications. Also, future systems should be much cheaper for consumers.

From Table 23.2, we can summarize the 4G concepts as following:

- 4G networks will provide subscribers with a higher bandwidth and a mobile data rate of 100 Mbps and more.
- It is expected that 3G networks will not be able to meet the needs of services like video-conferencing, full motion video, etc. in terms of QOS.
- There will be greater mobility and lower costs.
- It will be possible to integrate WLAN and WAN.

Table 23.2 Comparison of 3G and 4G network technologies

Feature	3G networks	4G networks
Driving force	Predominantly voice driven, data is secondary concern	Converged data and multimedia services over IP
Network architecture	Wide area cell based	Integration of wireless LAN and wide area networks
Frequency band	1.6–2.5 GHz	2–8 GHz
Data rate	385 Kbps–2 Mbps	20–100 Mbps and 1Gbps for stationary
Bandwidth	5 MHz	100+ MHz
Switching technique	Circuit and/Packet switched	Completely digital with packet voice
Radio access technology IP	CDMA family (WCDMA/CDMA2000) Multiple versions	OFDMA family (MC-CDMA or OFDM) IPv6.0
Major characteristic	Predominantly voice-data as add-on	Converged data and VoIP
Component design	Optimized antenna; multi-band adapters	Smart antennas; SW multi-band; wideband radios
Forward error correction	Convolution code 1/2, 1/3; turbo	Concatenated coding
Mobile top speed	200 kmph	200 kmph

The technology may see some peculiar features, such as cell phones operating in very high speed vehicles (e.g., trains running at more than 210 km/h). Present subscriber requirements include downloading videos and music etc., but the future seems to be moving towards applications like online games that demand immense capacity, greater QOS, and very low costs. 4G system must be capable of providing highly efficient and cost-effective solutions for wireless network users.

23.5 Applications of 4G

4G offers three-dimensional visual experiences. Thus, 4G will represent another quantum leap in mobile Internet speeds and picture quality.

4G will have better support of roaming and handoffs across heterogeneous networks. Users, even in today's wireless market, demand service transparency and roaming. 4G may support interoperability between disparate network technologies by using techniques such as LAS-CDMA signalling. Other solutions such as software-defined radios could also support roaming across disparate network technologies in 4G systems.

One of the most notable advanced applications for 4G systems is location-based services. 4G location applications would be based on visualized, virtual navigation schemes that would support a remote database containing graphical representations of streets, buildings, and other physical characteristics of a large metropolitan area. This database could be accessed by a subscriber in a moving vehicle equipped with the appropriate wireless device, which would provide the platform on which would appear a virtual representation of the environment ahead. For example, one would be able to see the internal layout of a building during an emergency rescue. This type of application is sometimes referred to as "Telegeoprocessing," which is a combination of Geographical Information Systems (GIS) and Global Positioning Systems (GPS) working in concert over a high-capacity wireless mobile system.

Telegeoprocessing over 4G networks will make it possible for the public safety community to have wireless operational functionality and specialized applications for everyday operations, as well as for crisis management. The emergence of next generation wireless technologies will enhance the effectiveness of the existing methods used by public safety. 3G technologies and beyond could possibly bring the following new features to public safety:

Virtual navigation: As described, a remote database contains the graphical representation of streets, buildings, and physical characteristics of a large metropolis. Blocks of this database are transmitted in rapid sequence to a vehicle, where a rendering program permits the occupants to visualize the environment ahead. They may also "virtually" see the internal layout of buildings to plan an emergency rescue, or to plan to engage hostile elements hidden in the building.

Tele-medicine: A paramedic assisting a victim of a traffic accident in a remote location could access medical records (e.g., x-rays) and establish a video conference so that a remotely based surgeon could provide "on-scene" assistance. In such a circumstance, the paramedic could relay the victim's vital information (recorded locally) back to the hospital in real time, for review by the surgeon.

Crisis-management applications: These arise, for example, as a result of natural disasters where the entire communications infrastructure is in disarray. In such circumstances, restoring communications quickly is essential. With wideband wireless mobile communications, both limited and complete communications capabilities, including Internet and video services, could be set up in a matter of hours. In comparison, it may take days or even weeks to re-establish communications capabilities when a wire line network is rendered inoperable.

23.9 Limitations of 4G

4G is still passing through research and, therefore, there are some problems that need to be fixed in order for the users to benefit from it fully. Still there are some limitations for 4G communications.

- Operating area is one of the major limitations. Although 2G networks are frequently used, still there are many areas not served. This drawback passes over into future generation.
- The extensive publicity to 3G networks nowadays gives unrealistic expectations in communications available at anytime, anywhere. The public should understand that high speed data communications will not be equivalent to the wired Internet.
- When 3G and 4G network services are deployed, there may be a great deal of disappointment and perceptions could become negative. If this were to happen, neither 3G nor 4G may realize its full potential.
- Cost is another limitation. The equipment required to implement a next generation network is still very expensive. Carriers and providers have to plan carefully to make sure that expenses are kept realistic.
- Pay-per-use model of services are currently being applicable in the Asian networks which are difficult to implement in United States, where the public is used to a service for free model (e.g., the Internet).